

SANKET

Indian Sign Language for Sarkari Clerks

Team Name	Beyond Words
Institution	Dr. Kiran & Pallavi Patel Global University (KPGU)
Yi Chapter / City	YUVA Chapter · Gujarat
Track	Public Services — <i>Indian Sign Language for All</i>
Problem Statement Title	Indian Sign Language for All (mainstream ISL via training, technology, incentives, policy)
Team Members & Roles	Rudra Khaire (Full-Stack Developer) · Pratiksha Jawale (Team Lead) · Mahi Panchal (Design & Research) · Suhani Pawar (Content & Curriculum) · Sheena Sharma (ISL Research & Documentation)
Date (Chapter-Level Round)	22 August 2026

3.1 Problem Understanding

15 marks

A deaf citizen walks into a municipal office in India and cannot be understood. The clerk does not know Indian Sign Language (ISL), no interpreter is on duty, and the fallback — writing on paper or a sympathetic third person — is slow, awkward and privacy-breaking. She waits, then leaves without her answer. This is not a rare event: it is the daily reality at thousands of government counters, and it amounts to exclusion from basic civic services.

The scale is national. The World Health Organization estimates **~63 million Indians live with significant hearing impairment** (WHO deafness estimates, still cited in 2024); the National Association of the Deaf counts ~18 million as deaf. Yet only **~5% of deaf children are in school**, and only **~26% of working-age deaf adults are employed** (Census of India 2011 — 73.9% are marginal workers). Meanwhile the public servants who sit across the counter — roughly **3.2 million central plus 7.3 million state government employees** (Ministry of Finance; Union Government statistics, 2023) — have almost no ISL capability. The **RPwD Act 2016** recognises ISL as a language of instruction, and ISLRTC, the Government of India’s recognised ISL authority, maintains a certified dictionary — but neither reaches the counter. The interpreter supply chain tells the same story: **fewer than 300 certified ISL interpreters exist in all of India** to serve ~18 million deaf citizens (ISLRTC; National Association of the Deaf) — a municipal counter has effectively no chance of one being present.

Root causes, not symptoms:

- **Frontline capacity gap** — training is school-centric (oralism, NPPCD screening); no mechanism puts ISL on the desk where service happens.
- **No distribution channel** — certified ISLRTC content exists but is not packaged for a clerk with 3 free minutes on an office PC.
- **No accountability** — no department is measured on whether counters are communication-accessible, so there is no incentive to act.
- **Perceived cost** — traditional courses are long and costly; the unit of training was never matched to the unit of work.

3.2 Proposed Solution

25 marks

Solution: Sanket — “From 30 days of training to 30 seconds of service.”

One-line description: Sanket turns an ordinary sarkari counter into a two-way ISL interpreter — a 30-second service turn that also trains every clerk into the habit.

Solution type: Product Technology Platform Hybrid

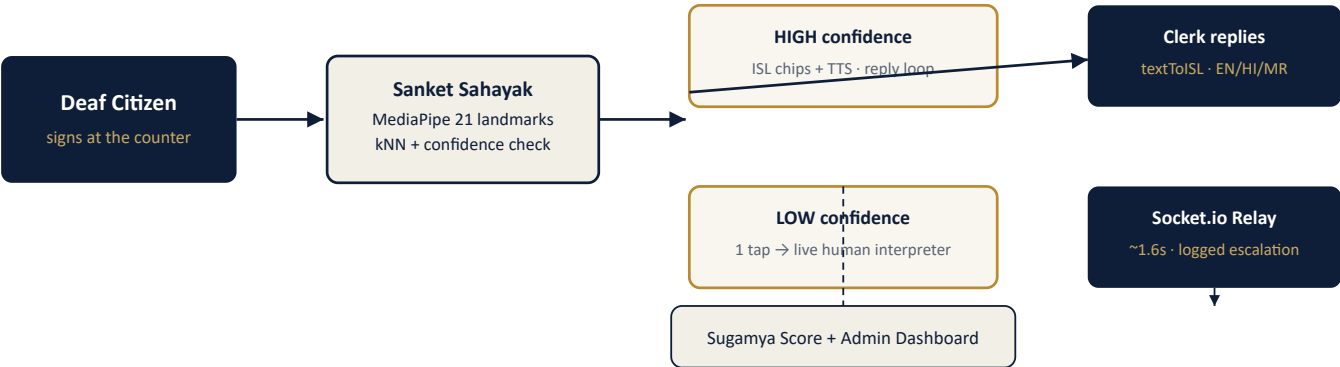
HOW IT WORKS

Sanket is a zero-install web platform with three pillars. **(1) Sanket Sahayak — service:** the citizen signs; a webcam running MediaPipe (21 hand landmarks) recognises the sign and shows the clerk text plus voice in English, Hindi or Marathi. The clerk replies in one tap and the reply is rendered as ISL symbol chips plus text-to-speech for the citizen — a complete service turn in under 30 seconds. When recognition confidence is low, one tap escalates to a **live human interpreter over a WebSocket relay**; the escalation is logged, not lost. **(2) ISL Quest — habit:** a 3-minute daily lesson (one video, one MCQ), five modules (Thank You, Please Wait, Sign Here, Water Bill, Submit) covering 34 high-frequency signs across six categories, with streaks, badges, XP and PDF certificates, plus 60-second webcam-based sign training (kNN classifier stored as a JSON file). **(3) Admin Oversight — governance:** a departmental **Sugamya Score** (45% compliance, 30% citizen satisfaction, 15% participation, 10% “human safety net” escalations handled), department leaderboards and QR citizen-feedback loop.

HOW IT ADDRESSES THE ROOT CAUSE

Sanket attacks all four root causes at once: it puts training *at the counter* (3-minute units on the same PC a clerk already has); it is the *distribution channel* ISLRTC content lacks (curriculum curated from ISLRTC with attribution); it creates *accountability* through the Sugamya Score; and its unit economics (one laptop, one weekend, free-tier stack) remove the cost barrier. Critically, it is honest about machine limits: recognition is a scaffold, and a human safety net guarantees the citizen is never blocked when the model is wrong.

Target users: government clerks and department admins (primary), deaf citizens at the counter (beneficiaries), and municipalities (purchasers).



Stack: Next.js · MongoDB Atlas (mock fallback) · PWA offline · runs in Chrome on existing office PCs

Fig. 1 — Sanket system architecture: service flow, escalation path, and governance loop.

3.3 Policy Recommendation

Feeds Solution Quality & Impact

Policy gap: The RPwD Act 2016 recognises ISL as a language of instruction and mandates digital/ICT accessibility (Section 42, operationalised through IS 17802 and the Accessible India — Sugamya Bharat — campaign), yet it imposes **no operational duty** on municipal departments to make service counters communication-accessible, and no metric exists to hold them accountable. A department can be fully “compliant” on paper while a deaf citizen still cannot pay a water bill.

Recommendation (specific policy ask):

1. **Mandate ISL-ready counters.** Amend the RPwD Rules / national accessibility standards to require that every public service counter designate and train at least one “ISL-ready clerk” per office, using a certified ISLRTC-sourced curriculum.
2. **Make accessibility measurable.** Require departments to report an annual **Counter Accessibility Score** (assisted citizens, escalations handled, citizen feedback) — the governance layer Sanket already produces.
3. **Routing the mandate:** a DoPT training circular adding a 3-minute daily ISL micro-module to existing government staff e-learning (myGov/iGOT), with ISLRTC as the content authority.
4. **Pilot trigger:** a directive piloting ISL-ready counters in one taluka office per district for 90 days before statewide rollout.

Acting bodies: Department of Empowerment of Persons with Disabilities (DEPwD), Ministry of Social Justice & Empowerment, jointly with the Department of Personnel & Training (DoPT) and ISLRTC.

How this scales Sanket nationally: a compliance duty creates procurement and adoption; a reported score creates accountability and a data market for the solution; ISLRTC-standardised content makes the training uniform across states, so Sanket’s curriculum becomes the reference implementation — from one taluka office to six lakh counters.

3.4 Feasibility & Scalability

20 marks

First 90 days (Pilot — one taluka office): Days 1–15: sign partnership with ISLRTC (content) and one municipal office; deploy on Vercel with MongoDB Atlas; train the pilot talati and two clerks (60-minute onboarding). Days 16–45: run the 30-second desk flow live; seed from ISLRTC curriculum; track assists and escalations daily. Days 46–75: QR citizen feedback loop live; first weekly department review using the Sugamya Score. Days 76–90: document assisted-citizen count and escalation log; prepare the municipal-to-state expansion brief. **Metric: assisted citizens per counter per month — not sign-ups.**

Rough cost estimate (pilot, INR):

- Infrastructure: free-tier Vercel + MongoDB Atlas (₹0); existing office PCs (₹0 new hardware).
- Onboarding & travel: ₹25,000.
- Content licensing/curation support: ₹50,000 (ISLRTC public content is open for educational use).
- Contingency & feedback material (QR stands, signage): ₹25,000.
- **Total ≈ ₹1.0–1.5 lakh for a 90-day single-office pilot** — per-assist, not per-licence economics; scaling adds server capacity only.

Infrastructure & partners: Chrome on existing office PCs; MediaPipe in-browser (no ML server); PWA offline for weak networks; mock fallback so the demo/service never crashes. Partners: ISLRTC (content), one municipality (pilot), deaf-community reviewers, local colleges (field support).

Key stakeholders: talati/clerks (users), department admins (governance), deaf citizens (beneficiaries), DEPwD/ISLRTC (policy + content), municipal IT (deployment).

Scalability roadmap: **Pilot** (1 taluka office, 90 days) → **City** (municipality-wide, 6–12 months, train-the-trainer champions) → **State** (district rollouts + adoption heatmaps, 12–24 months) → **National** (six lakh counters, SMS-first rural reach). Every phase is measured on assisted citizens, and each escalation log tells departments exactly where training must improve.

3.5 Innovation & Differentiation

15 marks

Most ISL efforts built a **recognizer**: a camera sees a sign and a screen shows text — and the service ends the moment the model is wrong. Sanket built the **routing**. The unique unlock is the **human-in-the-loop escalation**: when recognition confidence drops, a live human clerk who signs is one tap away over a WebSocket relay, and that escalation is logged and rewarded (10% of the Sugamya Score is “human safety net”). Recognition is treated honestly as a scaffold, not the product — and the content layer is anchored to ISLRTC’s certified **10,000-term dictionary** (3rd edition, 2021), not a private vocabulary.

Second, Sanket is **desk-first, not phone-first**: it is installed where service happens — the 30-second two-way desk — paired with a habit-building learning loop (3-minute lessons, streaks, certificates) that existing ISL apps (which teach vocabulary with no counter flow) do not combine. Third, it closes the **governance loop**: no competitor ships a departmental score that turns machine failure into a trainable, reportable number. Why not before? Because recognition was mistaken for the product; nobody shipped recognition + certified content + training + human escalation + accountability as one system that runs on hardware the government already owns.

3.6 Impact Potential

15 marks

Who benefits (quantified): the ~63M hearing-impaired Indians (WHO) — of whom ~18M are deaf (NAD) — gain access to basic civic services; ~10.5M central + state government employees gain an ISL capability path; the pilot office serves its deaf residents from day 1, with a conservative target of **50–100 assisted citizens in the first 90 days** in one taluka.

Measurable KPIs: assisted citizens per counter per month; escalations handled (and resolved); % clerks completing the 3-minute daily streak; department Sugamya Score; citizen satisfaction via QR feedback.

Timeframe to real impact: measurable assists within 30–60 days of the pilot; city-level impact in 6–12 months; state-level in 12–24 months. Impact is verifiable, not aspirational — every assist, escalation and feedback response is logged and visible to the department and to judges.

Annexures — References & Supporting Material

1. WHO — estimates of significant hearing impairment in India (~63M), deafness estimates. · 2. Census of India 2011 — disability statistics (50,71,007 deaf; 73.9% marginal workers among hearing-disabled aged 15–59). · 3. National Association of the Deaf (NAD) — ~18M deaf; interpreter shortage. · 4. ISLRTC (islrtc.nic.in) — Dept. of Empowerment of Persons with Disabilities, Ministry of Social Justice & Empowerment; established 28 Sept 2015; ISL Dictionary 1st ed. 2018 (3,000 terms), 2nd ed. 2019 (6,000), 3rd ed. 2021 (10,000); ISLRTC–NCERT MoU (6 Oct 2020); <300 certified interpreters. · 5. Rights of Persons with Disabilities (RPwD) Act, 2016 — ISL recognition; disabilities 7→21; reservation 3%→4%; Section 42 ICT accessibility; enacted Dec 2016, in force Apr 2017. · 6. Bureau of Indian Standards IS 17802 (Dec 2021) — ICT accessibility standard, enforceable via 2023 RPwD Rules amendment; Accessible India (Sugamya Bharat) campaign. · 7. Ministry of Finance — Annual Report on Pay & Allowances (central ~31.9 lakh); Union Government employee statistics 2023 (state ~73 lakh). · 8. NPPCD — National Programme for Prevention & Control of Deafness (2007 pilot, 587 districts by 2024). · 9. Yuva Future 6.0 problem statement — “Indian Sign Language for All”.

Supporting material: Sanket platform live demo (sanketsahayaks1.vercel.app), system architecture (Fig. 1), feature list, demo plan, Q&A bank — maintained in the project repository.

Sanket · Team Beyond Words · KPGU University — “From 30 days of training to 30 seconds of service.”